Software Project Management
Assignment-2
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following statements regarding the Waterfall model are correct?

a. Integration activities are distributed evenly throughout development.
b. Customer feedback is typically obtained through frequent prototypes.
c. Requirement changes after project initiation can be difficult to accommodate.
d. The model may create a false sense of project progress.
e. Working software may not be available until late in the project.

Correct Answer: c. Requirement changes after project initiation can be difficult to accommodate.
d. The model may create a false sense of project progress.
e. Working software may not be available until late in the project

Detailed Solution:

In the Waterfall model, requirements are fixed early, making late changes difficult and costly to accommodate. Since working software is delivered only at the end, progress is often measured by documents and completed phases, which can create a false sense of progress, and stakeholders do not see executable software until late in the project. Integration also happens at the end (not evenly distributed), and prototypes are not a standard part of Waterfall.

QUESTION 2:

Which of the following pairs are correctly matched in the V-Model of software development?

a. Requirements Specification ↔ System Testing
b. High-Level Design ↔ Integration Testing
c. Detailed Design ↔ Unit Testing
d. Project Planning ↔ Unit Testing
e. Coding ↔ System Testing

Correct Answer: a. Requirements Specification ↔ System Testing
b. High-Level Design ↔ Integration Testing
c. Detailed Design ↔ Unit Testing

Detailed Solution:

In the V-Model, each development phase on the left has a corresponding testing phase on the right: Requirements Specification maps to System Testing, High-Level Design maps to Integration Testing, and Detailed Design maps to Unit Testing. Project Planning has no direct testing counterpart, and Coding corresponds to Unit Testing—not System Testing.

QUESTION 3:

Which of the following are commonly cited reasons for using prototyping?

a. Guaranteed reduction in project duration.
b. Requirements are only partially known.
c. Improved communication between users and developers.
d. Elimination of all testing activities.
e. Improved user involvement.

Correct Answer: b. Requirements are only partially known.
c. Improved communication between users and developers
e. Improved user involvement

Detailed Solution:

Prototyping is commonly used when requirements are unclear or only partially known, as it helps clarify them through tangible feedback. It also improves communication between users and developers by giving users something concrete to interact with, and it actively involves users in the refinement process. However, prototyping does not guarantee shorter project duration (it can add time) and does not eliminate the need for testing.

QUESTION 4:

Which of the following are major objectives of the RAD model?

a. Reduce software development time.
b. Reduce software development cost.
c. Delay requirement changes until after testing.
d. Accommodate change requests as early as possible.
e. Eliminate the need for user involvement.

Correct Answer: a. Reduce software development time.
b. Reduce software development cost.
d. Accommodate change requests as early as possible

Detailed Solution:

The Rapid Application Development (RAD) model aims to reduce development time and cost through iterative prototyping, reusable components, and accelerated cycles. It also emphasizes accommodating change requests as early as possible, keeping the system flexible to user feedback. RAD does not delay changes until after testing—it welcomes early changes—and it relies heavily on continuous user involvement, not eliminates it.

QUESTION 5:

Which of the following statements about refactoring in the agile model are correct?

a. Refactoring requires changing user-visible behavior.
b. Refactoring improves internal code structure.
c. Refactoring should preserve external behavior.
d. Refactoring is intended to add major new functionality.
e. Refactoring can improve maintainability.

Correct Answer: b. Refactoring improves internal code structure.
c. Refactoring should preserve external behavior.
e. Refactoring can improve maintainability.

Detailed Solution:

Refactoring improves the internal structure of the code without altering its external behavior, which preserves functionality while making the system cleaner and easier to maintain. It is not meant to add new features or change what the user sees—those are separate development activities.

QUESTION 6:

Which of the following are important practices in the XP model of software development?

a. Frequent integration instead of end-of-project integration.
b. Ongoing refactoring and design improvement.
c. Strict requirement freeze before development starts.
d. Greater emphasis on feedback and adaptation.
e. Testing only after implementation is complete.

Correct Answer: a. Frequent integration instead of end-of-project integration.
b. Ongoing refactoring and design improvement.
d. Greater emphasis on feedback and adaptation

Detailed Solution:

Extreme Programming (XP) emphasizes frequent integration (often multiple times a day), continuous refactoring and design improvement to keep code clean, and strong feedback and adaptation through practices like test-driven development and customer involvement. XP does not freeze requirements early—it welcomes changing requirements—and it promotes testing before or alongside implementation, not only after.

QUESTION 7:

Which of the following statements are FALSE regarding Scrum Sprints?

a. A Sprint is a time-boxed development cycle.
b. Software increments can be designed, coded, and tested within a Sprint.
c. Sprint duration is exactly one month.
d. Changes are normally entertained throughout the Sprint without restriction.
e. Scrum projects progress through a series of Sprints.

Correct Answer: c. Sprint duration is exactly one month.
d. Changes are normally entertained throughout the Sprint without restriction

Detailed Solution:

Sprint duration is not fixed at exactly one month—it is typically 1 to 4 weeks, and the team chooses the length. Also, once a Sprint begins, the Sprint Backlog is fixed, and changes are not entertained without restriction; any change requires canceling the Sprint or deferring it to the next Sprint. The other statements (a, b, and e) are true regarding Scrum Sprints.

QUESTION 8:

Which of the following statements regarding the Product Backlog are correct?

a. It contains only completed tasks.
b. It contains a list of desired work for the project.
c. It may contain both story-based and task-based work items.
d. It is prioritized by the Product Owner.
e. It cannot be modified once the project starts.

Correct Answer: b. It contains a list of desired work for the project.
c. It may contain both story-based and task-based work items.
d. It is prioritized by the Product Owner.

Detailed Solution:

The Product Backlog contains a list of all desired work for the project, including user stories, technical tasks, and bug fixes, and it is continuously prioritized by the Product Owner to maximize value. It is not limited to completed tasks—those are removed or marked done—and it is dynamic, meaning it can be refined, reprioritized, or updated throughout the project, not frozen at the start.

QUESTION 9:

Which of the following are typically represented in a Sprint Burn Down Chart?

a. Customer satisfaction ratings.
b. Remaining effort or hours.
c. Progress across Sprint days.
d. Number of employees in the organization.
e. Work remaining until Sprint completion.

Correct Answer: b. Remaining effort or hours.
c. Progress across Sprint days
e. Work remaining until Sprint completion

Detailed Solution:

A Sprint Burndown Chart tracks the remaining effort (e.g., hours or story points) across the days of the Sprint, showing how much work is left until completion. It does not display customer satisfaction, organizational headcount, or any metric outside the Sprint's work progress.

QUESTION 10:

Which of the following statements about Product Backlog prioritization are correct?

a. Backlog items are typically ordered according to priority.
b. The Product Owner is responsible for backlog prioritization.
c. Prioritization helps determine what work should be addressed first.
d. All backlog items must have equal priority.
e. Priorities cannot change during a project.

Correct Answer: a. Backlog items are typically ordered according to priority.
b. The Product Owner is responsible for backlog prioritization.
c. Prioritization helps determine what work should be addressed first.

Detailed Solution:

The Product Backlog is an ordered list where items are ranked by priority, and the Product Owner is solely responsible for this prioritization to ensure the team works on the most valuable items first. Prioritization directly guides the sequence of development, but items do not have equal priority—they vary in importance—and priorities can and do change frequently based on feedback, market conditions, or business needs.

******THE END******